RDMA: Cost Effective Agent-Driven Rare Disease Mining from Electronic Health Records

John Wu[1*], Jimeng Sun[1] and Adam Cross[2]

[1*]Department of Computer Science, University of Illinois Urbana-Champaign, Champaign, IL, USA.
[2]Department of Pediatrics, University of Illinois College of Medicine Peoria, Peoria, IL, USA.

*Corresponding author(s). E-mail(s): johnwu3@illinois.edu;
Contributing authors: jimeng@illinois.edu; arcross@uic.edu;

Abstract

Rare diseases affect 1 in 10 Americans yet remain systematically underdocumented in clinical records. ICD-based systems cannot capture their breadth, over 50% of Orphanet codes lack a direct ICD mapping and only 2.2% of HPO codes have matching ICD codes, leaving patient populations invisible and delaying diagnosis. Mining unstructured clinical notes offers a direct path forward, but real notes are long, noisy, and abbreviation-dense, and limited annotations make fine-tuning infeasible, demanding approaches that generalize without task-specific training. We present Rare Disease Mining Agents (RDMA), an agentic framework equipping smaller quantized LLMs with tools for abbreviation resolution, implicit phenotype reasoning, and ontology grounding against Orphanet and HPO. RDMA substantially outperforms fine-tuned and RAG-based baselines across benchmarks with different data characteristics, without any task-specific training. A small quantized model achieves maximal performance, reducing inference costs by up to 10x and local hardware costs by up to 17x, enabling private deployment on standard hardware without cloud-based PHI exposure. RDMA's uncertainty-flagging mechanism further reduces expert annotation burden while preserving agreement quality, supporting scalable rare disease documentation in clinical practice. Available at https://github.com/jhnwu3/RDMA.

Keywords: Rare Disease, Agents, Data Mining

Introduction

Rare diseases affect approximately 1 in 10 Americans, constituting a significant healthcare challenge despite their individual scarcity [1]. Accurate diagnosis remains difficult due to the vast diversity and sparsity of these conditions [2]. While efforts to map international classification of diseases (ICD) codes to more granular Orphanet codes have been attempted [3], over 50% of Orpha codes lack a direct mapping, resulting in the systematic under-reporting of rare diseases in ICD-annotated systems. Furthermore, only 2.2% of HPO codes have matching ICD codes [4]. Existing rare disease differential diagnosis benchmarks rely directly on ICD labels without Orphanet or HPO concept annotations [5], potentially overstating real-world performance [6]. General purpose clinical NLP tools such as cTAKES [7] and MetaMap [8] inherit this same limitation, as they ground to ICD and UMLS terminologies, making automated documentation of rare disease conditions challenging. Mining clinical notes from EHRs offers a promising path forward, enabling direct mapping to HPO [9] and Orphanet [10] without relying on ICD codes as intermediaries [11, 12]. However, real clinical notes are long, noisy, and filled with abbreviations and implicit clinical reasoning, exposing three critical gaps in existing mining approaches.

First, existing public mining benchmarks are either poorly annotated or fail to reflect real-world clinical notes, making fine-tuning on them unreliable or infeasible. For example, our physicians found that annotations in earlier MIMIC-III rare disease mention mining attempts [11, 13] frequently misinterpreted clinical abbreviations, such as interpreting NPH as normal pressure hydrocephalus" rather than neutral protamine hagedorn" in insulin treatment contexts. Additionally, phenotype mining approaches like RAG-HPO [14] and PhenoGPT [12] evaluate on clinical case studies that lack the abbreviations and misspellings found in real clinical notes, and these case studies [12, 15] typically contain only several hundred words, whereas real clinical notes span thousands [16, 17]. Several studies [18, 19] further rely on private annotations, impeding reproducible development. As a key consequence, approaches trained on one benchmark often fail to generalize across benchmarks with different data characteristics, and fine-tuning for specific deployment settings is frequently unrealistic, demanding methods that are robust without any task-specific training.

Second, a substantial portion of clinically relevant phenotypes remain implicit rather than explicitly stated in notes. Identifying these requires interpreting laboratory results and applying clinical reasoning, tasks that go beyond simple entity extraction. Conventional approaches typically follow a two-stage extract-and-match pipeline that fails to capture these implied phenotypes. Most existing methods [14, 18, 19] first extract entities and then match or verify them against ontologies like HPO [9] or Orphanet [10] using retrieval-augmented generation (RAG) [20]. Dictionary-based approaches [21–23] like FastHPOCR [24], while extremely efficient, lack the ability to reason about implied entities or the broader context surrounding identified mentions.

Third, privacy concerns present significant barriers to deploying capable models in practice. While cloud services increasingly adopt HIPAA compliance measures [25, 26], deploying LLM APIs with protected health information (PHI) typically requires extensive institutional review board (IRB) scrutiny [27]. Locally-deployable solutions offer advantages in privacy protection and audit transparency [28], but local deployment has historically required large, expensive hardware.

These three challenges together define what an effective mining solution must provide: generalization across data characteristics without task-specific training, reasoning capabilities that go beyond surface-level extraction, and practical deployability under real-world privacy constraints. Large language models (LLMs) have emerged as natural candidates given their flexibility, strong performance across clinical tasks [29], and ability to leverage existing knowledge bases such as HPO and Orphanet without extensive retraining [14, 20, 30]. LLMs can also provide interpretability through explanations [31], making them well-suited for mining new phenotypes that require reasoning beyond semantic similarity. However, existing LLM-based approaches do not resolve the three challenges above: they are evaluated on private benchmarks [18, 19], rely on explicit entity extraction without reasoning about implied phenotypes [14, 18], and typically require cloud deployment of large models [14]. Critically, we find that simply scaling to larger LLMs does not overcome these reasoning limitations, making the design of the agentic workflow, rather than model size, the determining factor in robust performance.

To address these challenges, we propose Rare Disease Mining Agents (RDMA), an agentic framework that coordinates LLM-based reasoning across structured ontology access, implicit phenotype inference, and uncertainty-flagging for human review (Fig. 1). RDMA makes three key contributions: (1) it generalizes substantially better across benchmarks without any task-specific training, outperforming fine-tuned and RAG-based baselines across datasets with different data characteristics; (2) it achieves maximal performance with a small 4-bit quantized model, reducing inference costs by up to 10x and local hardware costs by up to 17x, making private deployment on standard consumer hardware (RTX 3090) practical; and (3) its uncertainty-flagging mechanism reduces expert annotation burden while preserving agreement quality, offering a blueprint for human-AI collaboration in refining noisy clinical datasets. We hope our exploration serves as a useful guide for building rigorous rare disease benchmarks and differential diagnosis frameworks.

Results

Experimental Setup. Clinical note mining for rare disease differential diagnosis involves two key tasks: extracting phenotypes and identifying rare diseases. Here, we summarize the experimental setup towards evaluating existing mining approaches and RDMA.

Datasets. Existing rare disease datasets have known limitations but remain useful benchmarks. Phenotype identification. We evaluate on the case study cohort (CSC) [14] and BioLark gold standard corpora [12]. The CSC benchmark better reflects clinical notes with more text and phenotypes per document, though it still falls short of real-world clinical complexity. Rare disease identification. Three benchmarks exist: RDD [32], RareDis [15], and a noisily annotated MIMIC-III disease benchmark [11, 13]. We exclude RDD as it lacks rare disease mention annotations. To demonstrate RDMA's ability to both improve extraction and assist annotators, we construct a gold

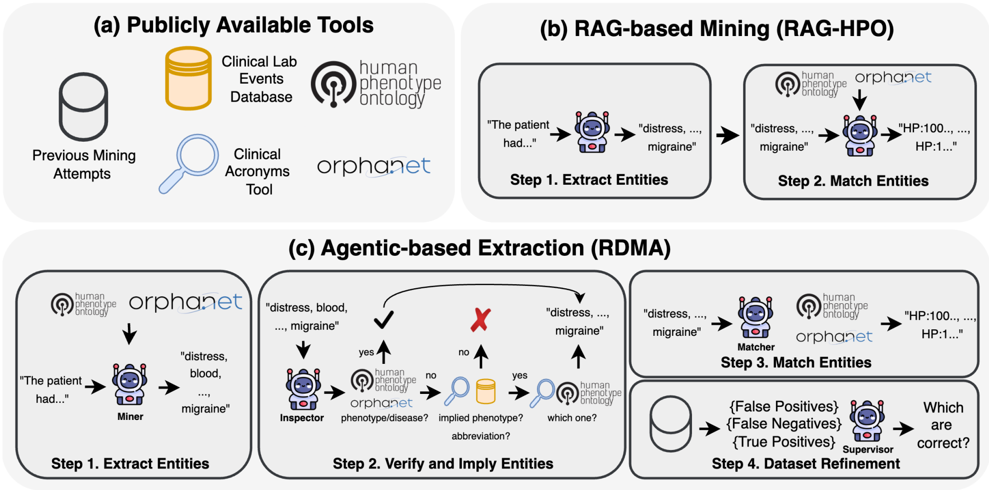

Fig. 1

evaluation set from the MIMIC-III annotations via combined human and agent annotation, yielding a realistic benchmark on real clinical notes. All benchmark statistics are described in Supplementary Table 6.

Evaluation. Our goal is to map text evidence to structured codes from the HPO [9] and Orphanet [10] ontologies, framing extraction as a medical coding task [33]. Phenotype extraction. Annotated codes are available for both phenotype datasets, making evaluation straightforward. Rare disease identification. RareDis provides only text span annotations without ontology mappings. As both exist in our MIMIC3 evaluation set, we report both span and code mapping performance here as MIMIC3-RD Entity and MIMIC3-RD Code respectively. For baselines lacking an ontology mapping step, we choose to use a fuzzy matching retrieval approach that is employed by the authors of PhenoGPT [12]. All evaluation metrics are described in the Supplementary Material.

Baselines. Across all benchmarks, approaches fall into three categories: (1) rule-based methods that tokenize text and match to ontologies via nearest-neighbor search, (2) fine-tuned encoder models trained to extract entities directly, and (3) LLM-based methods using zero-shot or RAG-based extract-and-match pipelines. For rule-based approaches, we include a simple retrieve-and-string-match Dictionary Match baseline using state-of-the-art clinical embeddings MedEmbed [34] and the state-of-the-art FastHPOCR. For fine-tuned encoders, we distinguish between off-the-shelf models and those we fine-tune ourselves. Off-the-shelf, we evaluate PhenoGPT [12] and Stanza's i2b2 Clinical BERT, both used without modification. Following the RareDis authors [35], we additionally fine-tune BioBERT [36] and BioClinicalBERT [37] for rare disease and phenotype extraction; all four encoder models are evaluated across all benchmark datasets. For LLM-based approaches, we test a range of models in both zero-shot and RAG-based settings [14]. We note that highly specialized approaches such as FastHPOCR and PhenoGPT do not apply to our rare disease extraction benchmarks.

Extraction Performance. We evaluate mining performance along two axes: generalization across distributions and scaling across model sizes in Table 1 and Fig. 2. Specifically, we ask (1) how well does each approach generalize across benchmarks with different data characteristics, and (2) how does performance scale with model size. We find that (1) our agentic system is substantially more robust than competing approaches, outperforming specialized baselines without requiring any training, and (2) a small quantized model is sufficient for maximal performance across all benchmarks, making RDMA computationally feasible compared to LLM-based approaches that depend on larger models for robust performance.

		Phenotype Mining						Rare Disease Mining								
		BiolarkGSC+			CSC			RareDis			MIMIC3-RD Entity			MIMIC3-RD Code		
Approach	**FT**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Non-LLM Baselines																
Dictionary Match	✗	0.682	0.214	0.326	0.600	0.210	0.310	0.850	0.410	0.550	0.400	0.468	0.431	0.300	0.450	0.360
FastHPOCR	✗	0.721	0.518	**0.603**	0.520	0.450	0.480		*N/A*			*N/A*			*N/A*	
i2b2 Clinical BERT	✓	0.599	0.417	0.491	0.480	0.600	0.530	0.153	0.850	0.260	0.010	0.879	0.020	0.010	0.651	0.019
BioBERT†	✓	0.635	0.409	0.498	0.614	0.278	0.382	0.730	0.703	0.716	0.018	0.559	0.033	0.025	0.473	0.047
BioClinicalBERT†	✓	0.459	0.424	0.441	0.449	0.348	0.392	0.681	0.666	0.673	0.015	0.473	0.027	0.026	0.527	0.050
LLM-based Approaches																
PhenoGPT	✓	0.676	0.312	0.427	0.570	0.390	0.460		*N/A*			*N/A*			*N/A*	
Zero-Shot (Llama 3.3 70B)	✗	0.000	0.000	0.000	0.000	0.000	0.000	0.900	0.570	0.700	0.141	0.602	0.228	0.001	0.007	0.002
RAG (RD / HPO) (Llama 3.3 70B)	✗	0.410	0.563	0.475	0.674	0.580	0.624	0.130	0.890	0.230	0.045	0.716	0.085	0.017	0.570	0.030
RDMA (Mistral 24B)*	✗	0.565	0.553	0.559	0.644	0.671	**0.657**	0.870	0.760	**0.810**	0.516	0.695	**0.592**	0.460	0.620	**0.530**

Table 1 Standardized Performance Comparison Across Phenotype and Rare Disease Mining Approaches. We evaluate both domains side-by-side using various LLM backbones for generative methods. For Rare Disease evaluation, BioBERT and BioClinicalBERT† are fine-tuned on datasets (BiolarkGSC+, RareDis) where span annotations exist and evaluated on their evaluation counterparts (CSC, MIMIC3-RD) here. "N/A" blocks denote models whose architectures or pre-training are not applicable to the respective dataset. Colors are normalized per benchmark so that the darkest orange always reflects the highest score within each column group. FT indicates whether the model was fine-tuned on any biomedical corpus before usage. **Bold** denotes best performance, underline denotes second best. We report our bootstrapped confidence intervals in Supplementary Tables 7 and 8.

Agentic LLM approaches better generalize across distributions without domain-specific training. For datasets where annotations do not contain explicit text spans and sample sizes are small, such as CSC and MIMIC3, fine-tuning is practically infeasible, and even where it is feasible, supervised models trained on one corpus frequently fail to generalize to out-of-distribution data. As shown in Table 1, fine-tuned models such as PhenoGPT, BioBERT, and BioClinicalBERT require extensive training on well-annotated corpora such as BiolarkGSC+ and RareDis, yet exhibit substantial performance degradation when evaluated on out-of-distribution benchmarks like CSC and MIMIC3. Rule-based approaches like FastHPOCR avoid this training requirement but exhibit similar brittleness: FastHPOCR [24] well on the benchmark it was originally evaluated on, but underperforms on CSC [14]. By contrast, RDMA generalizes across extraction tasks without any additional training, providing a substantial performance uplift across 4 of the 5 benchmarks compared to both trained and rule-based alternatives.

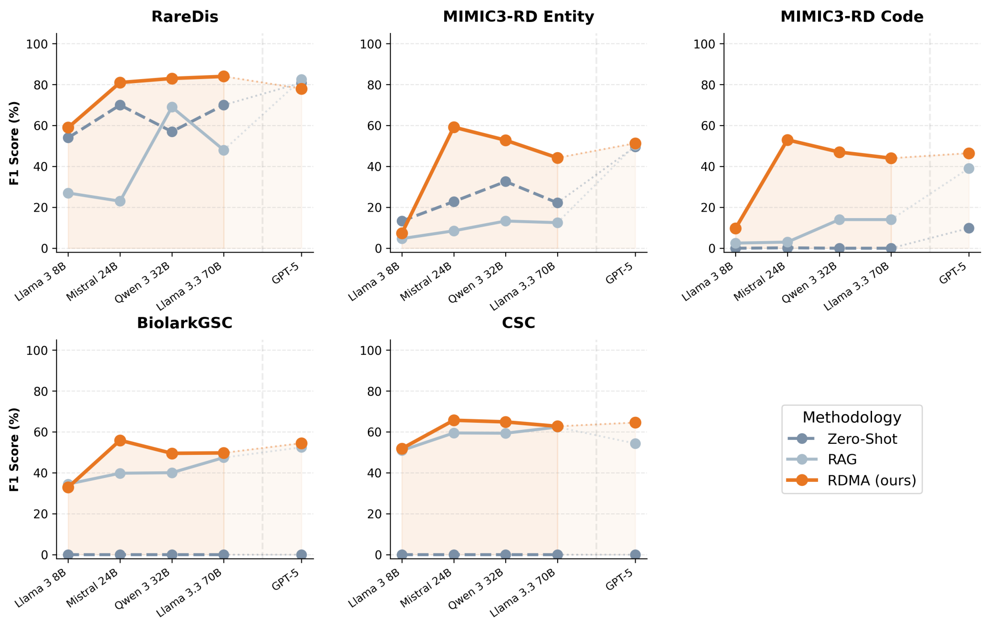

Fig. 2

LLMs hallucinate HPO and Orpha codes. In benchmarks requiring both entity extraction and ontology code assignment, we observe that LLMs frequently hallucinate invalid HPO and Orpha codes despite correctly identifying the underlying text spans, as evidenced by the performance gap between text-span and code-based benchmarks in Table 1. Providing retrieved ontology context in the RAG setting substantially improves code-level performance, but often at the cost of recall as false positives increase. We hypothesize that retrieved ontology context biases the model toward over-predicting entity presence, surfacing mentions that are not actually attested in the source text.

Larger models do not improve extraction performance. Beyond 8B models, which frequently struggle to follow structured instructions, we observe that increasing model size from 24B up to proprietary models such as GPT-5 yields no consistent improvement in rare disease or phenotype extraction performance across any benchmark, with gains effectively plateauing.

RDMA provides a more a stable foundation for LLM mining performance. Different LLMs can respond very differently to the same prompts, and as shown in Fig. 2, extraction performance varies substantially across model families in both zero-shot and RAG settings on the RareDis and MIMIC3-RD Entity benchmarks. Notably, RDMA consistently achieves the highest or comparable F1 across all configurations regardless of the underlying LLM.

Larger models are too expensive to deploy locally on large clinical corpuses. Estimating mining costs on MIMIC-IV notes [17] based on run time and GPU rental

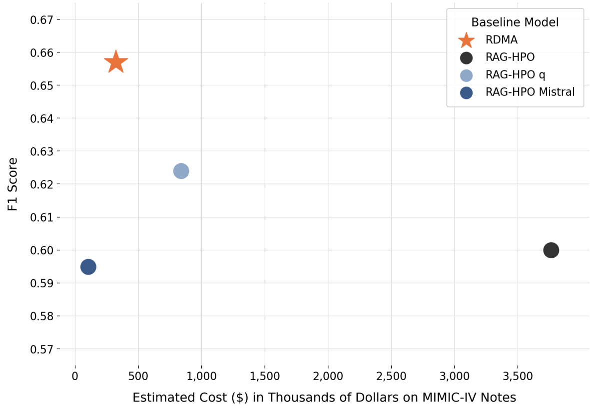

Fig. 3

prices, we observe that larger non-quantized LLMs are substantially more expensive to run (i.e up to 10x) than their smaller quantized counterparts in Fig. 3.

Additional reasoning steps enable smaller quantized models to surpass larger ones. While the performance gap between RAG-based approaches and RDMA narrows as model size increases in Fig. 2, smaller models remain substantially cheaper to run. As illustrated in Fig. 3, our quantized 24B Mistral model exceeds the performance of the much larger non-quantized Llama 3.3 70B at a fraction of the cost. Moreover, as shown in Fig. 2, RDMA performance is comparable to our cloud-based GPT-5 baseline regardless of the underlying framework, suggesting that frontier models do not necessarily confer an advantage on rare disease mining tasks.

Agent-Assisted Annotations Case Study. We use our cleaning of the MIMIC3-RD annotations as a case study to explore how agentic systems can contribute beyond robust entity mining—specifically, by reducing annotator burden while preserving annotation quality. We compare two correction strategies: one in which a human annotator reviews all annotations, and one in which the annotator reviews only those flagged by the agent.

Accelerating annotation workflows. RDMA supports clinical experts by surfacing retrieved candidate entities, contextual information, and previously annotated Orphacodes in a streamlined format, directly addressing the time constraints faced by busy clinicians.

To focus expert attention where it matters most, RDMA employs a two-stage verification process. First, it compares its mining results against prior annotations to compute pseudo false negatives, false positives, and true positives. Second, a verifier module flags the most contentious cases: false negatives that are not rare diseases, false positives that are rare diseases, and true positives that are not rare diseases, precisely the cases where human judgment is most valuable.

This targeted review reduces annotation burden by 63%, decreasing the number of cases requiring re-review from 333 to 122 (Table 2), while maintaining high agreement with fully human-annotated references (Table 3). As a qualitative example, Supplementary Fig. 1 shows RDMA correctly identifying that “NPH” had been mislabeled as a rare disease; the surrounding context instead pointed to insulin resistance, a correction subsequently confirmed by human experts.

RDMA identifies overlooked disease mentions. Beyond reducing workload, RDMA also improves annotation completeness. As shown in Table 2, RDMA-assisted annotation uncovers a greater number of unique rare diseases than human annotation alone, with at least 10 rare disease mentions present in the clinical notes going undetected during initial human review. This highlights RDMA's dual role: as a validation tool that audits existing annotations, and as a discovery mechanism that recovers entities missed in preliminary efforts. Qualitative examples of such overlooked entities are provided in Supplementary Figs. 2 and 3.

Taken together, these results demonstrate that agentic systems can simultaneously reduce annotator workload and improve the accuracy and completeness of the resulting annotations. For more information, annotation guidelines are detailed in Supplementary Fig. 4.

Metric	Initial	Human Only	RDMA & Human
Total Docs	117	117	117
Annotations Re-reviewed	N/A	**333**	**122**
Unique Rare Diseases	192	**120**	**135**

Table 2 Dataset comparison across annotation stages. Human correction reduces unique rare diseases from 192 to 120 (−37.5%), likely removing false positives. RDMA & Human recovers annotations to 135 (+12.5% from human-only), capturing valid rare diseases missed initially. **RDMA reduces annotations requiring human re-review by over 63%.**

Metric	RDMA Only	RDMA & Human
Cohen's Kappa	0.46	0.81
F1 Score	0.74	0.94
Precision	0.92	0.92
Recall	0.62	0.96

Table 3 Human in the Loop Refinement via Two Stage Discrepancy Audit. In step 4 of RDMA, the agent acts as a tie breaker analyzing entity sets from two separate agentic passes, computing pseudo FP, FN, and TP labels before auditing its own classifications to flag high-uncertainty cases for human review. After final human review, agreement with expert annotators remains high, suggesting that the flagging accurately filters for the most important entities needing re-review.

Discussion

The RDMA framework addresses critical bottlenecks in clinical information extraction by leveraging the adaptability of agentic workflows. This section explores the strengths of our approach, the persistent challenges of medical reasoning, and the need for more robust benchmarking in rare disease research.

The flexibility of agentic frameworks is a key advantage over fixed pipelines, owing to their task-specific adaptability. As shown in Fig. 1, phenotype and disease extraction agents utilize distinct tools while maintaining a shared workflow of extract, verify,

Baseline	F1	P	R
RDMA	**0.66**	0.64	0.67
No Lab Tool	0.64	0.63	0.64
No Impl. Check	0.65	**0.64**	0.65
No Verif.	0.53	0.42	**0.71**

Table 4 RDMA Ablation Study. Impact of individual components on phenotype mining performance of a quantized Mistral 24B model on the CSC [14].

match, and refine. RDMA allows for modular customization—such as including abbreviation detection or lab event databases for phenotype extraction while omitting lab events for rare disease extraction, where they provide no value. Our ablation study (Table 4) underscores the importance of this modularity. While the lab test tool and implication reasoning provide marginal gains in recall, the verification step is the most critical component, providing a substantial boost to precision by filtering out hallucinations and misalignment.

Despite the strengths of agentic systems, significant challenges remain in phenotype mining and medical reasoning, particularly in capturing implied phenotypes. Analysis of our benchmark (Fig. 4) reveals that over 25% of phenotypes are not explicitly mentioned in the text. These require LLMs to infer conditions from lab results or symptom clusters. Current LLMs remain insufficient for basic reasoning tasks like implying phenotypes related to lab events, as shown by the tiny improvements in Fig. 4. RDMA still struggles with: (1) long-context reasoning: failing to link distal mentions (e.g., a colonoscopy performed earlier in a note to a "colonic tubular adenoma"); (2) terminology nuance: identifying "decreased visual acuity" as a match for the HPO code "reduced visual acuity"; and (3) negation detection: occasionally ignoring negative qualifiers (e.g., "negative for..."), even when prompted to consider negations.

Robustness to clinical note complexity is essential: a primary hurdle in rare disease mining is the discrepancy between scientific passages and real-world EHRs. Clinical notes in datasets like MIMIC-IV are often thousands of words long and laden with noise such as abbreviations and misspellings. As shown in Fig. 5, RDMA demonstrates remarkable robustness to document length. Unlike one-shot RAG approaches that typically perform a single extraction and degrade as context grows, RDMA's recall slightly increases with longer texts. By retrieving against individual sentences and employing agentic verification, we maintain a stable F1 score across varying lengths. Nonetheless, a great area of interest is improving the ability of these systems at longer and noisier contexts, especially given that the performance discrepancy of approaches on the scientifc RareDis corpus and our cleaned MIMIC3 clinical note corpus has a gap of over 20% F1 as shown in Table 1.

There is a pressing need for comprehensive rare disease benchmarks. Current datasets for mining [14, 38, 39] and differential diagnosis (DDx) [19, 40, 41] face significant limitations. Sentence snippets and scientific passages may not reflect the lengthy, noisy nature of EHRs. Furthermore, many DDx benchmarks use ICD codes as ground-truth proxies, which are flawed because they do not capture rare diseases precisely enough and are often missing from the text [42, 43]. Expecting LLMs to perform accurate DDx without the ability to extract all phenotypes reliably is premature. RDMA provides a framework to generate more accurate ORPHA and HPO-based profiles, but the field requires larger, expert-verified datasets that encompass the full diversity of the Orphanet ontology [10] and HPO [9]. For future work, extending our mining approach with human verification on the entire MIMIC-III [13] and MIMIC-IV [44] corpus would be ideal. Gold-standard rare disease and phenotype annotations do not currently exist for MIMIC-IV at scale, however, making this a critical next step.

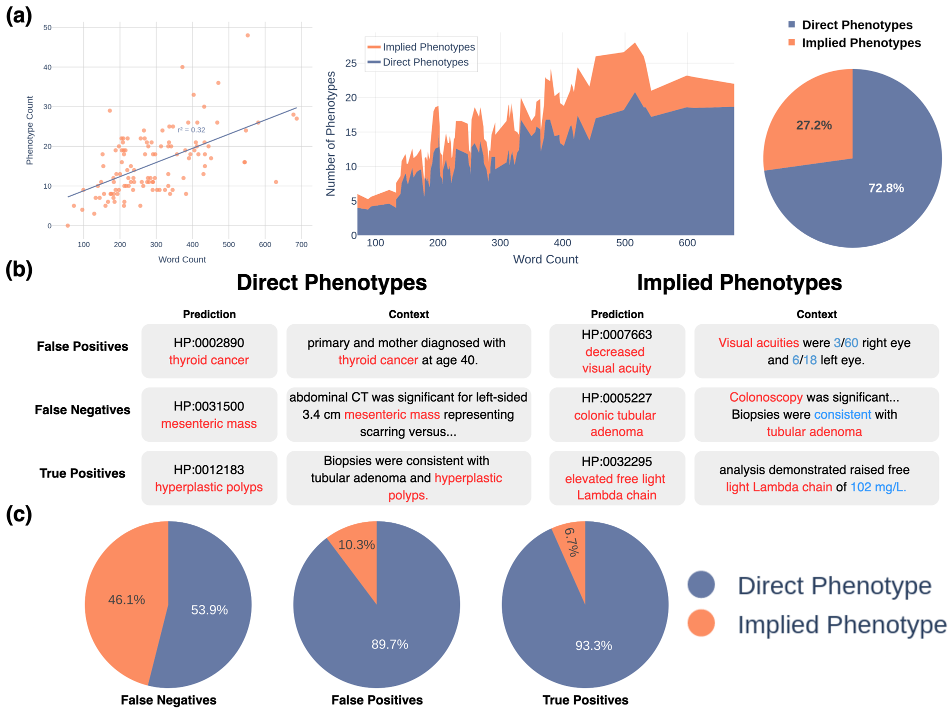

Fig. 4

Methodology

Problem Formulation. RDMA operates over a corpus of clinical documents and aims to identify two types of entities within each document: phenotypes, represented as HPO codes from the Human Phenotype Ontology [9], and rare diseases, represented as ORPHA codes from the Orphanet ontology [10]. The output is an annotated dataset where each document is paired with the set of phenotypes and rare diseases found within it.

Tool Retrieval. Most of RDMA's tools are structured as retrieval systems over medical databases. Given a candidate entity string, we embed it and retrieve the most semantically similar entries from the relevant ontology database using Euclidean distance over embeddings. We use FAISS [45] for efficient search and MedEmbed Small [34] as our embedding model, chosen for its strong performance on medical text.

RDMA Overview. Algorithm 1 presents the full RDMA pipeline. At a high level, RDMA processes each clinical document sentence by sentence, using semantic search to retrieve ontology candidates and an LLM to extract entities of interest. Each candidate entity then passes through a multi-step verification pipeline before being mapped to a formal ontological code. Finally, the resulting annotations are compared against a prior annotation set, with disagreements flagged for human expert review. Not all steps apply to every entity type, phenotype extraction involves lab event

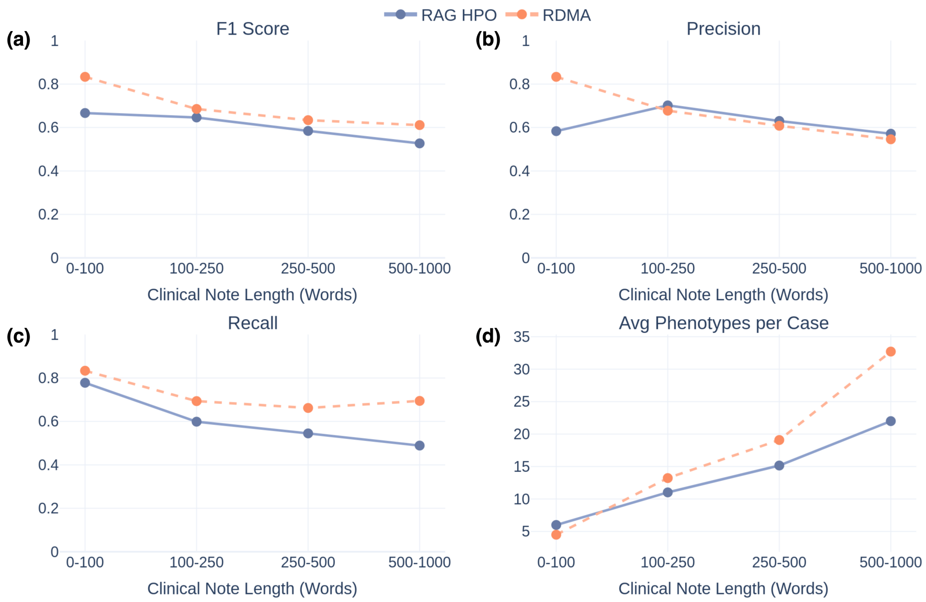

Fig. 5

detection and implied phenotype generation, while rare disease extraction focuses on abbreviation expansion and confirming that matched entries are true diseases rather than related biological entities. We describe each step below.

Algorithm 1 RDMA: Mining Pipeline

Require: Corpus X, ontologies $\mathcal{O} = \mathcal{D}_{HPO} \cup \mathcal{D}_{Orphanet}$, mode $m \in \{\text{rare}, \text{phenotype}, \text{both}\}$

Ensure: Annotated dataset D

```
for each document x ∈ X do
    for each sentence s in x do
        cands_s ← SemanticSearch(s, O)
        E ← LLMExtract(s, cands_s)
    end for
    for each entity e ∈ E do
        if m ∈ {rare, both} then
            e ← ExpandAbbreviation(e)
            if OntologyMatch(e, s) and LLMIsDisease(e) then
                Accept e as verified rare disease
            end if
        end if
        if m ∈ {phenotype, both} then
            if OntologyMatch(e, s) then
                Accept e as verified phenotype
            else if IsAbnormalLabValue(e) then
                Accept LLMGenerateDirection(e) as implied phenotype
            else if LLMImpliesPhenotype(e) is valid in D_HPO then
                Accept implied phenotype
            end if
        end if
    end for
    for each verified entity e do
        cands_e ← SemanticSearch(e, O)
        code ← LLMMatch(e, s, cands_e)
    end for
end for
return D
```

Algorithm 2 RDMA: Dataset Refinement

Require: D, prior annotations $\mathcal{A}_{prior}$

Ensure: Flagged entities $\mathcal{F}$ for review

```
A_prior ← KeywordFilter(A_prior)
TP, FN, FP ← Compare(D, A_prior)
for each e ∈ TP ∪ FN ∪ FP do
    v ← OntologyMatch(e)
    if (e ∈ TP and not v) or then
       (e ∈ FN and v) or
       (e ∈ FP and v)
        F ← F ∪ {e}
    end if
end for
return F
```

Entity Extraction. Each document is split into sentences, and for each sentence we query the relevant ontology database to retrieve the top-5 most similar phenotype or disease candidates. Both the sentence and its candidates are passed to an LLM via LLMExtract, which identifies any phenotype or rare disease entities worth investigating further. Sentence-level chunking balances specificity against computational cost: finer chunks yield more precise retrieval but increase processing time, a trade-off we examine further in Supplementary Table 9.

Entity Verification and Implication. Each extracted entity passes through a verification pipeline before being accepted. The steps differ for phenotypes versus rare diseases (see general structure in Algorithm 1, lines 7–20).

Abbreviation expansion. For rare disease candidates, we first attempt to expand any clinical abbreviations via ExpandAbbreviation before proceeding, since abbreviation ambiguity is particularly problematic in this setting.

Direct ontology verification. We call OntologyMatch to check whether the entity genuinely corresponds to an HPO or Orphanet entry in context. For rare diseases, we additionally call LLMIsDisease to confirm the matched entry is a disease rather than a related entity type (e.g., a gene, protein, or enzyme) that also appears in Orphanet.

Lab event detection and implication (phenotypes only). If a phenotype entity is not directly verified, we check whether it describes an abnormal lab measurement via IsAbnormalLabValue. If so, LLMGenerateDirection converts it into the corresponding implied phenotype (e.g., elevated creatinine → hypercreatininemia).

Implied phenotype generation (phenotypes only). For entities that are neither direct phenotypes nor lab events, LLMImpliesPhenotype checks whether the entity contextually implies a phenotype. If so, we generate and verify the implied phenotype against the HPO ontology before accepting it.

Verified Entity Matching. Each verified entity is mapped to a specific ontological code by LLMMatch, which is presented with the entity, its source sentence, and the top retrieved ontology candidates. This follows a similar workflow to RAG-HPO [14], extended here to cover rare disease identification.

Dataset Refinement. After assembling mined annotations, RDMA compares them against a prior annotation set using Algorithm 2. Entities that agree across both sources are treated as likely correct; disagreements are flagged via FlagForReview for human expert judgment. The flagging logic is asymmetric by design: a true positive that fails re-verification, a false negative that passes, or a false positive that passes are all suspicious and warrant review, while cases where both sources consistently agree or disagree are left unflagged.

Preliminary Filtering. Before running RDMA refinement, we apply a set of rule-based corrections via KeywordFilter, informed by a clinician review of the prior annotations (Supplementary Table 10). This removed over 700 misannotated entries — for example, "MR" had been tagged as "Multicentric reticulohistiocytosis" when it almost universally refers to magnetic resonance in context — reducing the dataset from 1,073 annotations across 312 notes to 333 annotations across 117 notes (Supplementary Table 11). The full set of LLM prompts used by RDMA is provided in Supplementary Figs. 5–16.

Ethics. This study was approved under institutional review board (IRB) protocol 1994008-8, a joint approval of the University of Illinois College of Medicine Peoria and the University of Illinois Urbana-Champaign. All datasets analysed were publicly sourced and de-identified, and use of the MIMIC-III database complies with the PhysioNet Credentialed Health Data License 1.5.0. All human annotation and evaluation were performed by the co-authors with their consent.

Acknowledgements. This study was funded by Jump ARCHES endowment awarded by the Healthcare Engineering Systems Center at the University of Illinois Urbana-Champaign (UIUC) and Order of Saint Francis (OSF) Foundation. The funder played no role in study design, data collection, analysis and interpretation of data, or the writing of this manuscript.

Declarations

- Competing interests: The authors declare no competing financial or non-financial interests.
- Data availability: The datasets generated and/or analysed during the current study are available in the RDMA repository at https://github.com/jhnwu3/RDMA.
- Code availability: The underlying code for this study is available in the RDMA repository and can be accessed via this link https://github.com/jhnwu3/RDMA.
- Author contributions: JW wrote the manuscript, designed the figures, and implemented the experiments. AC and JS participated in the design, analysis, and discussion of the experiments. AC and JS also assisted with revisions to the manuscript. All authors read and approved the final manuscript.

.

References

[1] Virginia Tech: One in 10 Americans Is Living with a Rare Disease. Virginia Tech News. Accessed: 2025-04-02 (2025). news.vt.edu/articles/2025/02/research_fralinbiomed_rarediseaseday2025_0228.html

[2] Auvin, S., Irwin, J., Abi-Aad, P., Battersby, A.: The problem of rarity: estimation of prevalence in rare disease. Value in Health **21**(5), 501–507 (2018)

[3] Cavero-Carbonell, C., Rico, J., Garibay, L., García-López, M., Guardiola-Vilarroig, S., Maceda-Roldán, L., Zurriaga, O.: From icd10 to orphacodes: paving the way towards improved identification systems for rare diseases. European Journal of Public Health **30**(Supplement_5), 166–494 (2020)

[4] Tan, A.L., Gonçalves, R.S., Yuan, W., Brat, G.A., Gentleman, R., Kohane, I.S.: Implications of mappings between international classification of diseases clinical diagnosis codes and human phenotype ontology terms. JAMIA open **7**(4), 118 (2024)

[5] Zhao, W., Wu, C., Fan, Y., Zhang, X., Qiu, P., Sun, Y., Zhou, X., Wang, Y., Sun, X., Zhang, Y., Yu, Y., Sun, K., Xie, W.: An Agentic System for Rare Disease Diagnosis with Traceable Reasoning (2026). https://arxiv.org/abs/2506.20430

[6] Nguyen, M.-H., Yang, C.-T., Cassini, T.A., Ma, F., Hamid, R., Bastarache, L., Peterson, J.F., Xu, H., Li, L., Ma, S., et al.: Diagnostic accuracy of large language models for rare diseases: A systematic review and meta-analysis. medRxiv, 2026–03 (2026)

[7] Wiegreffe, S., Choi, E., Yan, S., Sun, J., Eisenstein, J.: Clinical concept extraction for document-level coding. In: Proceedings of the 18th BioNLP Workshop and Shared Task, pp. 261–272 (2019)

[8] Khin, N.P.P., Lynn, K.T.: Medical concept extraction: A comparison of statistical and semantic methods. In: 2017 18th IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD), pp. 35–38 (2017). IEEE

[9] Gargano, M.A., Matentzoglu, N., Coleman, B., Addo-Lartey, E.B., Anagnostopoulos, A.V., Anderton, J., Avillach, P., Bagley, A.M., Bakštein, E., Balhoff, J.P., *et al.*: ¡? mode longauthoraffil?¿ the human phenotype ontology in 2024: phenotypes around the world. Nucleic acids research **52**(D1), 1333–1346 (2024)

[10] Weinreich, S.S., Mangon, R., Sikkens, J., Teeuw, M.E., Cornel, M.: Orphanet: a european database for rare diseases. Nederlands tijdschrift voor geneeskunde **152**(9), 518–519 (2008)

[11] Dong, H., Suárez-Paniagua, V., Zhang, H., Wang, M., Casey, A., Davidson, E., Chen, J., Alex, B., Whiteley, W., Wu, H.: Ontology-driven and weakly supervised rare disease identification from clinical notes. BMC Medical Informatics and Decision Making **23**(1), 86 (2023)

[12] Yang, J., Liu, C., Deng, W., Wu, D., Weng, C., Zhou, Y., Wang, K.: Enhancing Phenotype Recognition in Clinical Notes Using Large Language Models: PhenoBCBERT and PhenoGPT (2023). https://arxiv.org/abs/2308.06294

[13] Johnson, A.E., Pollard, T.J., Shen, L., Lehman, L.-w.H., Feng, M., Ghassemi, M., Moody, B., Szolovits, P., Anthony Celi, L., Mark, R.G.: Mimic-iii, a freely accessible critical care database. Scientific data **3**(1), 1–9 (2016)

[14] Garcia, B.T., Westerfield, L., Yelemali, P., Gogate, N., Rivera-Munoz, E.A., Du, H., Dawood, M., Jolly, A., Lupski, J.R., Posey, J.E.: Improving automated deep phenotyping through large language models using retrieval augmented generation. medRxiv, 2024–12 (2024)

[15] Martínez-deMiguel, C., Segura-Bedmar, I., Chacón-Solano, E., Guerrero-Aspizua, S.: The raredis corpus: a corpus annotated with rare diseases, their signs and

symptoms. Journal of biomedical informatics **125**, 103961 (2022)

[16] Edin, J., Junge, A., Havtorn, J.D., Borgholt, L., Maistro, M., Ruotsalo, T., Maaløe, L.: Automated medical coding on mimic-iii and mimic-iv: a critical review and replicability study. In: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 2572–2582 (2023)

[17] Johnson, A., et al.: MIMIC-IV-Note: Deidentified free-text clinical notes. PhysioNet (2023). https://doi.org/10.13026/1n74-ne17 . https://doi.org/10.13026/1n74-ne17

[18] Wu, J., Dong, H., Li, Z., Wang, H., Li, R., Patra, A., Dai, C., Ali, W., Scordis, P., Wu, H.: A hybrid framework with large language models for rare disease phenotyping. BMC Medical Informatics and Decision Making **24**(1), 289 (2024)

[19] Chen, X., Mao, X., Guo, Q., Wang, L., Zhang, S., Chen, T.: RareBench: Can LLMs Serve as Rare Diseases Specialists? (2024). https://arxiv.org/abs/2402.06341

[20] Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., Wang, H.: Retrieval-Augmented Generation for Large Language Models: A Survey (2024). https://arxiv.org/abs/2312.10997

[21] Deisseroth, C.A., Birgmeier, J., Bodle, E.E., Kohler, J.N., Matalon, D.R., Nazarenko, Y., Genetti, C.A., Brownstein, C.A., Schmitz-Abe, K., Schoch, K., *et al.*: Clinphen extracts and prioritizes patient phenotypes directly from medical records to expedite genetic disease diagnosis. Genetics in Medicine **21**(7), 1585–1593 (2019)

[22] Liu, C., Peres Kury, F.S., Li, Z., Ta, C., Wang, K., Weng, C.: Doc2hpo: a web application for efficient and accurate hpo concept curation. Nucleic acids research **47**(W1), 566–570 (2019)

[23] Yadav, V., Bethard, S.: A survey on recent advances in named entity recognition from deep learning models. In: Proceedings of the 27th International Conference on Computational Linguistics, pp. 2145–2158 (2018)

[24] Groza, T., Gration, D., Baynam, G., Robinson, P.N.: Fasthpocr: pragmatic, fast, and accurate concept recognition using the human phenotype ontology. Bioinformatics **40**(7), 406 (2024)

[25] Ayad, M., Rodriguez, H., Squire, J.: Addressing hipaa security and privacy requirements in the microsoft cloud. Windows% 20Azure% 20HIPAA% 20Implementation% 20Guidance. pdf (2011)

[26] Keshetti, S., *et al.*: Designing scalable and hipaa-compliant notification systems

for healthcare: Leveraging cloud, microservices, and secure architectures. In: International Journal for Research Publication and Seminar, vol. 16, pp. 154–173 (2025)

[27] Grady, C.: Institutional review boards: Purpose and challenges. Chest **148**(5), 1148–1155 (2015)

[28] Sun, Q., Wu, H., Zhang, X.S.: On Active Privacy Auditing in Supervised Fine-tuning for White-Box Language Models (2024). https://arxiv.org/abs/2411.07070

[29] Wang, C., Li, M., He, J., Wang, Z., Darzi, E., Chen, Z., Ye, J., Li, T., Su, Y., Ke, J., Qu, K., Li, S., Yu, Y., Liò, P., Wang, T., Wang, Y.G., Shen, Y.: A Survey for Large Language Models in Biomedicine (2024). https://arxiv.org/abs/2409.00133

[30] Sanmartin, D.: Kg-rag: Bridging the gap between knowledge and creativity. arXiv preprint arXiv:2405.12035 (2024)

[31] Savage, T., Nayak, A., Gallo, R., Rangan, E., Chen, J.H.: Diagnostic reasoning prompts reveal the potential for large language model interpretability in medicine. NPJ Digital Medicine **7**(1), 20 (2024)

[32] Fabregat, H., Araujo, L., Martinez-Romo, J.: Deep neural models for extracting entities and relationships in the new rdd corpus relating disabilities and rare diseases. Computer methods and programs in biomedicine **164**, 121–129 (2018)

[33] Edin, J., Junge, A., Havtorn, J.D., Borgholt, L., Maistro, M., Ruotsalo, T., Maaløe, L.: Automated medical coding on mimic-iii and mimic-iv: A critical review and replicability study. In: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval. SIGIR '23, pp. 2572–2582. ACM, ??? (2023). https://doi.org/10.1145/3539618.3591918 . http://dx.doi.org/10.1145/3539618.3591918

[34] Balachandran, A.: MedEmbed: Medical-Focused Embedding Models. https://github.com/abhinand5/MedEmbed

[35] Segura-Bedmar, I., Camino-Perdones, D., Guerrero-Aspizua, S.: Exploring deep learning methods for recognizing rare diseases and their clinical manifestations from texts. BMC bioinformatics **23**(1), 263 (2022)

[36] Sun, C., Yang, Z., Wang, L., Zhang, Y., Lin, H., Wang, J.: Biomedical named entity recognition using bert in the machine reading comprehension framework. Journal of Biomedical Informatics **118**, 103799 (2021)

[37] Alsentzer, E., Murphy, J.R., Boag, W., Weng, W.-H., Jin, D., Naumann, T., McDermott, M.B.A.: Publicly Available Clinical BERT Embeddings (2019).

https://arxiv.org/abs/1904.03323

[38] Lobo, M., Lamurias, A., Couto, F.M.: Identifying human phenotype terms by combining machine learning and validation rules. BioMed Research International **2017**(1), 8565739 (2017)

[39] Martínez-deMiguel, C., Segura-Bedmar, I., Chacón-Solano, E., Guerrero-Aspizua, S.: The RareDis corpus: a corpus annotated with rare diseases, their signs and symptoms (2021). https://arxiv.org/abs/2108.01204

[40] Chen, X., Jin, Y., Mao, X., Wang, L., Zhang, S., Chen, T.: RareAgents: Advancing Rare Disease Care through LLM-Empowered Multi-disciplinary Team (2025). https://arxiv.org/abs/2412.12475

[41] Germain, D.P., Gruson, D., Malcles, M., Garcelon, N.: Applying artificial intelligence to rare diseases: a literature review highlighting lessons from fabry disease. Orphanet Journal of Rare Diseases **20**, 186 (2025)

[42] Mazzucato, M., Pozza, L.V.D., Facchin, P., Angin, C., Agius, F., Cavero-Carbonell, C., Corrochano, V., Hanusova, K., Kirch, K., Lambert, D., *et al.*: Orphacodes use for the coding of rare diseases: comparison of the accuracy and cross country comparability. Orphanet Journal of Rare Diseases **18**(1), 267 (2023)

[43] Cheng, H., Jafari, R., Russell, A., Klopfer, R., Lu, E., Striner, B., Gormley, M.: MDACE: MIMIC documents annotated with code evidence. In: Rogers, A., Boyd-Graber, J., Okazaki, N. (eds.) Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 7534–7550. Association for Computational Linguistics, Toronto, Canada (2023). https://doi.org/10.18653/v1/2023.acl-long.416 . https://aclanthology.org/2023.acl-long.416/

[44] Johnson, A.E., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T.J., Hao, S., Moody, B., Gow, B., *et al.*: Mimic-iv, a freely accessible electronic health record dataset. Scientific data **10**(1), 1 (2023)

[45] Douze, M., Guzhva, A., Deng, C., Johnson, J., Szilvasy, G., Mazaré, P.-E., Lomeli, M., Hosseini, L., Jégou, H.: The Faiss library (2025). https://arxiv.org/abs/2401.08281

[46] Wu, H., Toti, G., Morley, K.I., Ibrahim, Z.M., Folarin, A., Jackson, R., Kartoglu, I., Agrawal, A., Stringer, C., Gale, D., *et al.*: Semehr: A general-purpose semantic search system to surface semantic data from clinical notes for tailored care, trial recruitment, and clinical research. Journal of the American Medical Informatics Association **25**(5), 530–537 (2018)

[47] Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., Chen, Z., Tang, J., Chen, X., Lin, Y., *et al.*: A survey on large language model based autonomous agents. Frontiers of Computer Science **18**(6), 186345 (2024)

[48] Pricing, S.: Pricing (2025). https://salad.com/pricing

[49] Hyperstack: GPU Pricing (2025). https://www.hyperstack.cloud/gpu-pricing

Figure Legends

Fig. 1 RDMA vs. RAG-based approaches. (b) RAG-based approaches [12, 14, 18] typically follow a two-stage pipeline: first extracting entities using specialized extractors like SemEHR [18, 46], general-purpose LLMs [14], or finetuned LLMs [12], then matching these entities to HPO codes [12, 18, 46]. (c) Our RDMA approach extends this paradigm to an agentic framework [47], doubling the number of tools in our extraction pipeline (a), incorporating additional reasoning steps for verification and implication detection. Specifically, armed with lab event guidelines and a clinical acronyms database, the agent is forced to reason about whether an extracted entity implies a specific phenotype or rare disease. Furthermore, similar to how annotators re-review uncertain annotations, agents are required to perform a dataset refinement step to highlight challenging and uncertain cases by comparing previous mining attempts.

Fig. 2 Performance of LLM rare disease mining performance across model sizes and approaches. We observe that in general increasing the model sizes does not improve performance of downstream mining approaches for rare diseases beyond a certain model size and intelligence, namely the Mistral 24B model. We also observe that mining performance for baseline RAG or zero-shot approaches fluctuate dramatically as seen on the RareDis and MIMIC3-RD Entity benchmarks.

Fig. 3 Phenotype mining inference costs with LLMs. RDMA uses a 4-bit quantized Mistral 24B model yet outperforms RAG-HPO variants running on both a 4-bit and full non-quantized Llama 3.3 70B, models that are substantially more expensive to run locally. Although RDMA's additional reasoning steps incur a modest overhead when normalized by model size, this cost is far smaller than the penalty of scaling to larger models. We use GPU rental pricing from cloud-providers [48] [49] to compute our estimated inference costs in Supplementary Table 4.

Fig. 4 CSC Phenotype Extraction Qualitative Analysis. We evaluate on 116 clinical cases (33,824 words total) from [14]. **(a) Dataset Statistics.** Document length weakly correlates with phenotype count. Of 1,813 total phenotypes, 1,320 (72.8%) appear explicitly while 493 (27.2%) are implied. **(b) Qualitative Analysis.** Our method captures phenotypes implied by lab tests. Key challenges include: (1) False positives that are clinically valid but lack exact HPO matches; (2) False negatives arising from missing non-obvious entities or failing to capture long-range context. RDMA also identifies direct phenotypes like "thyroid cancer" that human annotators missed. **(c) Performance Breakdown.** Implied phenotypes comprise only 27.2% of labels but account for a disproportionate share of false negatives.

Fig. 5 RDMA outperforms local RAG-HPO across note lengths. This analysis is conducted on the CSC benchmark [14]. Real-world clinical notes (MIMIC-IV) are substantially longer than annotated case studies (median 1,320 words vs 271.5 words). While overall precision slightly decreases as note length increases (b), RDMA maintains stable recall performance (c), enabling it to outperform its RAG-based counterpart across all note lengths (a). Note count (d) shows phenotype count typically increases with document length.